THE UPPER SHRIEK CONSTRUCTIONS FORM A PSEUDO FUNCTOR ON SCHEMES OF FINITE TYPE OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0ATY (sources/stacks-duality/pinned-original-source.tex line 4370).

Allowed standard prerequisites (editorial).

- The arrow (g o f)^! -> f^! o g^! and its well definedness. Same chapter, Lemmas labelled lemma-shriek-well-defined and lemma-upper-shriek-composition, cited at proof lines 4378-4380. Supplies the transformations whose coherence is asserted.
- The base change map for a cartesian square and the two composition rules. Same chapter, equation labelled equation-sheafy, Lemma labelled lemma-compose-base-change-maps and Lemma labelled lemma-compose-base-change-maps-horizontal, cited at proof lines 4410-4422. Identify the arrows of the square with the gamma maps and compose them.
- The commuting of distant squares and the definition of a pseudo functor. Same chapter, Remark labelled remark-going-around, cited at proof lines 4420-4424, and Categories, Definition labelled categories-definition-functor-into-2-category, invoked in the statement. Give the last commutation gamma_A o gamma_C = gamma_C o gamma_A and the meaning of the assertion.

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AA0: sources/stacks-duality/pinned-original-source.tex, source lines 3986–3986. Same-chapter pointer into the author environment lemma at source line 3986; invoked at proof line None of the counted proof.

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It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ATX: sources/stacks-duality/pinned-original-source.tex, source lines 4161–4161. Same-chapter pointer into the author environment lemma at source line 4161; invoked at proof line 4411 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 003N: sources/stacks-chow/tags.txt, source lines 141–141. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original theorem 0F41: sources/stacks-chow/tags.txt, source lines 18886–18886. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4394 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9Z: sources/stacks-chow/tags.txt, source lines 12967–12967. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4395 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 0A9L: sources/stacks-duality/pinned-original-source.tex, source lines 988–988. Same-chapter pointer into the author environment equation at source line 988; invoked at proof line 4413 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ATQ: sources/stacks-duality/pinned-original-source.tex, source lines 1238–1238. Same-chapter pointer into the author environment lemma at source line 1238; invoked at proof line 4415 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ATR: sources/stacks-duality/pinned-original-source.tex, source lines 1375–1375. Same-chapter pointer into the author environment lemma at source line 1375; invoked at proof line 4416 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0ATS: sources/stacks-duality/pinned-original-source.tex, source lines 1471–1471. Same-chapter pointer into the author environment remark at source line 1471; invoked at proof line 4423 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0F42 the constructions of Lemmas 0AA0 and 0ATX define a pseudo functor from the category FTS_S into the 2-category of categories (see Categories, Definition 003N).*

Proof. To show this we have to prove given morphisms $f : X \to Y$, $g : Y \to Z$, $h : Z \to T$ that

$$\begin{array}{ccc} (h \circ g \circ f)^! & \xrightarrow[\gamma_{A+B}]{} & f^! \circ (h \circ g)^! \\ \downarrow{\scriptstyle \gamma_{B+C}} & & \downarrow{\scriptstyle \gamma_C} \\ (g \circ f)^! \circ h^! & \xrightarrow{\gamma_A} & f^! \circ g^! \circ h^! \end{array}$$

is commutative (for the meaning of the γ's, see below). To do this we choose a compactification $\overline{Z}$ of Z over T, then a compactification $\overline{Y}$ of Y over $\overline{Z}$, and then a compactification $\overline{X}$ of X over $\overline{Y}$. This uses More on Flatness, Theorem 0F41 and Lemma 0A9Z. Let $W \subset \overline{Y}$ be the inverse image of Z under $\overline{Y} \to \overline{Z}$ and let $U \subset V \subset \overline{X}$ be the inverse images of $Y \subset W$ under $\overline{X} \to \overline{Y}$. This produces the following diagram

$$\begin{array}{ccccccc} X & \longrightarrow & U & \longrightarrow & V & \longrightarrow & \overline{X} \\ \downarrow{\scriptstyle f} & & \downarrow & A & \downarrow & B & \downarrow \\ Y & \longrightarrow & Y & \longrightarrow & W & \longrightarrow & \overline{Y} \\ \downarrow{\scriptstyle g} & & \downarrow & & \downarrow & C & \downarrow \\ Z & \longrightarrow & Z & \longrightarrow & Z & \longrightarrow & \overline{Z} \\ \downarrow{\scriptstyle h} & & \downarrow & & \downarrow & & \downarrow \\ T & \longrightarrow & T & \longrightarrow & T & \longrightarrow & T \end{array}$$

Without introducing tons of notation but arguing exactly as in the proof of Lemma 0ATX we see that the maps in the first displayed diagram use the maps (0A9L) for the rectangles $A + B$, $B + C$, A, and C as indicated. Since by Lemmas 0ATQ and 0ATR we have $\gamma_{A+B} = \gamma_A \circ \gamma_B$ and $\gamma_{B+C} = \gamma_C \circ \gamma_B$ we conclude that the desired equality holds provided $\gamma_A \circ \gamma_C = \gamma_C \circ \gamma_A$. This is true because the two squares A and C only intersect in one point (similar to the last argument in Remark 0ATS). □